

Soumik Pramanik

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Github

Summary

Experienced Data Scientist and AI Engineer with 3+ years of experience designing and building scalable end-to-end data and machine learning pipelines for large-scale production systems. Skilled in Python, SQL, PySpark, TensorFlow, PyTorch, Docker, and cloud-based analytics platforms, with expertise spanning data engineering, machine learning, MLOps, and AI system development. First author of an international **Springer Nature Switzerland** on a novel computer vision dataset, covering image classification, feature extraction, and bias mitigation techniques along with future extension of image segmentation and detection. Strong background in transforming raw data into production-ready analytics and AI solutions, from data ingestion and feature engineering to model deployment.

Experience

Autocomplete Search suggestion for Rakuten Music.

Jan 2024

Data Scientist, Rakuten – Tokyo, JP

- Design a query summary framework to collect and analyze key statistics per query, processing large-scale raw user search logs and interaction data and enabling the generation of candidate suggestions for an Autocomplete service.
- Designed and implemented data collection pipelines (Python, Hadoop, PySpark) to extract essential features for training machine learning models, optimizing the ranking and relevance of Autocomplete suggestions.
- Developed and validated ranking models using machine learning techniques to improve relevance and ordering of autocomplete suggestions, the ranking algorithm has been **improved by 17%**, resulting in a **39% enhancement in overall performance** compared to the previous results.

Technologies: Python, SQL, Hadoop, PySpark, Kafka, Scikit-learn, TensorFlow

GSM Analyzer.

- Developed an advanced call analytics tool to monitor and analyze call failures, call updates, and end-to-end (E2E) call connection issues, providing actionable insights to improve system reliability.
- Implemented a scoring-based ranking system to prioritize failure codes by traffic impact and KPI degradation, enabling operations teams to focus on the most business-critical issues first.
- Enabled faster root-cause analysis and decision-making, reducing time-to-resolution for high-impact network incidents and supporting continuous improvement of network performance.

Technologies: BigQuery, Power Automate, Airflow, Python, DOMO

Autonomous Multi-Agent AI Maestro.

- Architected and led the design and development of an autonomous multi-agent AI platform, defining agent orchestration, governance, and scalable deployment patterns for AI-driven analytics and decision automation.
- Designed scalable data pipelines to collect, transform, and serve structured and unstructured data and store processed datasets in centralized databases for downstream AI agents and analytical workflows.
- Designed end-to-end AI workflows integrating RAG, dynamic SQL generation, agentic data science tools and automated insight generation across heterogeneous data sources, ensuring traceability, reliability, and production readiness.
- Established MLOps and lifecycle standards (API services, CI/CD, monitoring, failure handling) to deliver secure, scalable, and continuously operated AI services.

Technologies: Python, RAG, FastAPI, MCP, Docker, Airflow, GCP BigQuery, LlamaIndex, Langgraph, Agno

Publications

Fish92: A Novel Dataset for Indigenous Fish Classification

Dec 2024

Soumik Pramanik, Shrihari A, Prithwijit Guha

9th International conference on Computer vision & Image processing (CVIP), IIITDM Kancheepuram, India, 2024

Cham: Springer Nature Switzerland, 2025

doi.org/10.1007/978-3-031-93694-4_9

Highlights: Fish92 Dataset, Feature Extraction, Classification Models, Debiasing Techniques

Education

IIT Guwahati - MTech in Robotics and Artificial Intelligence

Sept 2021 – May 2023

- GPA: 8.60/10.0

• **Coursework:** Image processing, Machine Learning, Computer Vision, Database Management Systems

Govt. College of Engineering and Leather Technology - BTech in Computer

Aug 2015 – May 2019

Science

- GPA: 8.45/10.0

• **Coursework:** Data Structure, Algorithm, Operating Systems, Computer Networks

Skills/Technologies

Languages: C++, C, Python, SQL

Frameworks/Tools: Langgraph, Agno, Pyspark, Scikit-learn, JAX, TensorFlow, PyTorch

MLOps/Tools: FastAPI, Docker, Airflow, Git, GCP

Database: postgresql, ClickHouse

Libraries: Geopandas, Shapely, Fiona, OpenCv, Torch-vision

Achievements

- Awarded **Best Performer** in RM QCC 2025 (1H) for the project: *IMS Call Setup RCA*.
- Achieved **Top 1% (99th percentile)** in GATE 2021; qualified GATE consecutively in 2019, 2020, and 2021.